DPP	DEPARTMENTAL POLICY AND PROCEDURE		
Title	**SURFACTANT ADMINISTRATION**		
Department	**Neonatology Department**		
Applies to	**All NICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-NICU-023 (v2)**		
Last Version Approval Date	**28/1/2022**	Number of Pages	**5**

	NAME	POSITION	SIGNATURE	DATE
Prepared by:	**MS. JOICY JOY**	NICU Quality Representative		27/2/2024
	MS. AISHA ZAIBI SAAD AL-SHUMRANI	Neonatal Intensive Care Unit, Head Nurse		27/2/2024
	DR. TAYSEER HASSAN	Neonatology Resident		27/02/2024
Reviewed by:	**DRA. FAWZIA MOHAMMED FARAHAT**	Head, Neonatology and Pediatric Department		14/03/24
	MR. MANSOUR AHMED AL-ZAHRANI	Head, Quality Management and Patient Safety		14.3.24
Approved by:	**MR. SULTAN ALI AL-GAMDI**	Nursing Director		19-3-24
	DR. WAEL KAMEL ARBA	HED Assistant for Medical Affairs		19/3/24

APPROVAL DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
19/3 / 2024	7 / 4 / 2024	TRIENNIAL	18 / 3 / 2027

Baha_Cluster
Healthcluster-baha@moh.gov.sa
مستشفى المخواة العام
Al-Mikhwah General Hospital

1.0 PURPOSE

1.1. To provide guidelines for nurses' responsibility during surfactant administration for infant.

1.2. To improve infants' oxygenation and prevent complication.

2.0 DEFINITION/ABBREVIATION

2.1. **Surfactant**: is a surface active agent. It is a mixture of phospholipids that is secreted in the pulmonary alveoli and reduces the surface tension of pulmonary fluids, thus contributing to elastic properties of pulmonary tissues. It is administered via endotracheal tube into the infant's trachea as treatment of Respiratory Distress Syndrome (RDS) or hyaline membrane disease (HMD).

3.0 RESPONSIBILITY

3.1. NICU doctor

3.2. Staff nurses

4.0 POLICY

4.1. Surfactant or Survanta administration is indicated for treatment and prophylaxis of respiratory distress syndrome (RDS) or hyaline membrane disease (HMD) in premature infants. It is administered via endotracheal tube into the infant's trachea under strict aseptic technique.

4.2. Surfactant therapy must be administered by a physician under strict sterile technique, clearly written in the doctor's order and medication sheet, with the dose and route, date and time with signature and stamp of the ordering physician.

4.3. The nurse and respiratory therapy should assist the physician during administration of surfactant.

4.4. Strict aseptic technique and the ten rights of administering medications must be observed during surfactant administration.

4.5. Surfactant should be warmed at room temperature before administration, but never use artificial warming method.

4.6. The nurse assisting in the administration of Surfactant medication to the infant shall establish patient's identification, confirmed with the patient's medical record and Identification band.

4.7. Medications shall be checked with the physician's order, and if there is doubt, must be clarified before administration to ensure accuracy. Staff Nurse assigned to the patient should counter check the physician's order.

4.8. Newborn should be closely monitored throughout the procedure, if becomes hypoxic during surfactant administration, dosing should be stopped until the newborn is stable.

4.9. Crash cart, with intubation equipment, oxygen source and suction must be ready in case cardiopulmonary resuscitation is needed.

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5.0 PROCEDURE (S)

5.1. Verify the order for surfactant in the physician's order.

5.2. Confirm doctor's order and observed ten rights of drug administration. Warm the Surfactant to room temperature at least 20 minutes. Do not use artificial warming method.

5.3. Survanta should be inspected visually for discoloration, must be off- white to light brown.

5.4. Place infant in a radiant warmer and regulate temperature control accordingly to maintain normal body temperature and prevent hypothermia.

5.5. Identify correct infant by checking identification bracelet with medical record number, comparing name on medication sheet.

5.6. Position the newborn supine and ensure leads are connected to the bedside cardiopulmonary monitor to continuously monitor the vital signs.

5.7. Attach the infant to pulse oximeter to provide continuous or intermittent non-invasive method of determining oxygen saturation.

5.8. Wear gloves and other PPE needed to facilitate sterility of procedure.

5.9. Assist the physician in the administration of Survanta intratracheally using a 5 french end-hole catheter (shortened according to length of endotracheal tube (ETT) attached to a syringe pre-filled with the medicine.

5.10. The physician will insert the catheter into the endotracheal tube to just above the level of the carina. Follow the manufacturer's recommendation for administering the surfactant.

5.11. In premature infants less than 1250 grams' birth weight, preferably within 15 minutes after birth as preventive treatment.

5.12. To treat infants with RDS confirmed by x-ray and requiring mechanical ventilation, Survanta should be given preferably by 6 hours of age as a rescue dose.

5.13. Observe the newborn carefully and pay close attention to chest expansion, changes in tidal volume, need for increase / decrease in supplemental oxygen and alterations in vital signs, including pulse oximeter.

5.14. Suction secretions prior to administration of Survanta. Secretions interfere with gas flow predispose infant to obstruction of the passages including endotracheal tube.

5.15. Increase the ventilator setting according to physician's order during procedure to maintain appropriate oxygenation.

5.16. Survanta should be administered slowly in divided doses. Ventilate the infant in between the procedure to prevent cyanosis and to provide adequate air exchange and chest wall expansion.

5.17. Between doses of surfactant the newborn may be hand ventilated or placed on the ventilator. Closely observe for changes in respiratory function. To prevent cyanosis and to provide adequate air exchange and chest wall expansion.

5.18. After administration is complete, monitor vital signs and respiratory status very closely. Some newborn will require rapid changes in ventilator parameters. Once surfactant is absorbed, there is usually increase in respiratory compliance that requires adjustment of ventilator setting to decrease MAP and prevent over inflation or Hyperoximia.

5.19. Obtain arterial / capillary blood gases at least one hour after the procedure or as ordered to evaluate acid-base ventilation and oxygenation

5.20. Assist in chest x-ray procedure as ordered to assess patient's response to therapy. Revision of care plan might be needed in severe cases of Hyaline Membrane Disease.

5.21. Suction secretions 2 to 4 hours after the procedure. Suctioning is delayed to allow maximum effects of the medicine.

5.22. Observe the newborn and assess vital signs and possible complications or deterioration and refer to the physician for immediate intervention.

5.23. Documentation:

5.23.1. Time, type of surfactant and amount administered to the neonate, and the name of the physician who administered the surfactant.

5.23.2. Vital signs before and after the surfactant.

5.23.3. Changes in supplemental oxygen and ventilator parameters.

5.23.4. Respiratory assessment of newborn's work of breathing after surfactant and after changes in ventilator parameters / supplemental oxygen.

5.23.5. Newborn's tolerance of the procedure.

6.0 EQUIPMENTS

6.1. Surfactant warmed to room temperature
6.2. 10 ml syringe, 20-gauge needle
6.3. Bedside cardiac monitor
6.4. Sterile gloves, Sterile gown
6.5. Stethoscope
6.6. Sterile feeding tube 5-6.5 Fr
6.7. Ambu bag / ventilator
6.8. Suction equipment
6.9. Pulse oximeter
6.10. Oxygen source

7.0 ATTACHMENT/FORMS

7.1. Medication Administration Record (GDOH-NUR-MAR-191)
7.2. Vital Signs Chart (GDOH-NUR-VSC-193)
7.3. Physicians Order Form (GDOH-INP-POF-079)
7.4. Nurses Progress Notes (GDOH-NUR-NPN-203)

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8.0 REFERENCES

8.1. John Kattwinkel, (2006). Neonatal Resuscitation Text Book. (5th Ed.). American Academy of Pediatrics and American Heart Association.

8.2. Carole Kenner, Judy Wright Lott, (2007). Comprehensive Neonatal Nursing. (3th Ed.) Saunders.

8.3. Mhairi G. MacDonald, Jayashree Ramasethu, (2007) Atlas of Procedure in Neonatology (3rd Ed.). Lippincott Williams & Wilkins.

8.4. John P.Coherty, Eric C. Eichenwald, Ann R. Stark. (2008). Manual of Neonatal Care. (6th Ed.). Wolters Kluwer/ Lippincott Williams & Wilkins

8.5. Tricia Lacy Gomella, (2009). Neonatology Management Procedures, On-Call Problems, Diseases & Drugs. (4th Ed.). Appleton & Lange.

8.6. Guidelines for Neonatal Care, (NICU) MOH Saudi Arabia

9.0 REVISION CLAUSE

9.1. Modification of policy format, lay out and policy coding

KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

MEDICATION ADMINISTRATION RECORD

Admission date:	Weight:	Height:

ALLERGIES: ❒ NKA ❒ Yes (If yes, include allergy to medications, food & blood products):

Nurse's Notes:

1. When a new prescription sheet is used, all current treatments must be fully re-written, re-signed.
2. Check for entries in every section in order to avoid omissions.
3. Follow medication administration and transcription order policy
4. In the event of a medicine not given, enter the appropriate code in the administration box available in page 2

ONCE ONLY PRESCRIPTION (STAT)

Date	Time Ordered	Drug	Dose	Route	Time given	Assigned Nurse Initial	Staff Nurse Initial

INTRAVENOUS SOLUTION

Date	Time Ordered	Types/ Volume of Fluids	Types/ Amount of Additives	Rate	Starting time	Assigned Nurse Initial	Staff Nurse Initial	Date/Time Discontinued

NOTE: Utilize the last page if the above spaces provided are insufficient.

NURSES IDENTIFICATION:

Name	Job Number	Initial	Name	Job Number	Initial	Name	Job Number	Initial

0 000000 001915

SN

Name:_____ الاسم:	MRN: رقم الملف الطبي:

NON- ADMINISTRATION
If a dose is not administered by a nurse for any reason, the nurse should initial the administration record and enter the appropriate code number.

CODES:
1 = Allergic Reaction
2 = Patient Fasting
3 = Omitted for Clinical Reasons
4 = Patient Refused
5 = Patient Unavailability
6 = Drug Unavailability
7 = Self Administration

Medication Name:			**DATE TIME**							
Dose	Route	Frequency								
Initial dose :	Starting date:									
RN Name:	RN Initial:									
COUNTERSIGNED BY: (STAFF NURSE)										
Name:	Initial:									
Time : Date:	Other instructions:									
Time drug discontinued: Date: / /										

Medication Name:			**DATE TIME**							
Dose	Route	Frequency								
Initial dose:	Starting date:									
RN Name:	RN Initial:									
COUNTERSIGNED BY: (STAFF NURSE)										
Name:	Initial:									
Time : Date:	Other instructions:									
Time drug discontinued: Date: / /										

Medication Name:			**DATE TIME**							
Dose	Route	Frequency								
Initial dose:	Starting date:									
RN Name:	RN Initial:									
COUNTERSIGNED BY: (STAFF NURSE)										
Name:	Initial:									
Time : Date:	Other instructions:									
Time drug discontinued: Date: / /										

Medication Name:			**DATE TIME**							
Dose	Route	Frequency								
Initial dose:	Starting date:									
RN Name:	RN Initial:									
COUNTERSIGNED BY: (STAFF NURSE)										
Name:	Initial:									
Time : Date:	Other instructions:									
Time drug discontinued: Date: / /										

Medication Name:			**DATE TIME**							
Dose	Route	Frequency								
Initial dose:	Starting date:									
RN Name:	RN Initial:									
COUNTERSIGNED BY: (STAFF NURSE)										
Name:	Initial:									
Time : Date:	Other instructions:									
Time drug discontinued: Date: / /										



SN

Name: _____ الاسم:	MRN: رقم الملف الطبي:

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
COUNTERSIGNED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
COUNTERSIGNED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
COUNTERSIGNED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
COUNTERSIGNED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
COUNTERSIGNED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Name: _____ الاسم:	MRN: رقم الملف الطبي:

HIGH ALERT MEDICATION

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
DOUBLE CHECKED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
DOUBLE CHECKED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
DOUBLE CHECKED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
DOUBLE CHECKED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
DOUBLE CHECKED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

0 000000 001915 SN

Name:_____ الاسم:	MRN: رقم الملف الطبي:

INTRAVENOUS SOLUTION

Date	Time Ordered	Types/ Volume of Fluids	Types/ Amount of Additives	Rate	Starting time	Assigned Nurse Initial	Staff Nurse Initial	Date/Time

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MRN: |_|_|_|_|_|_|_|_|_|_| رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

VITAL SIGNS CHART

WEIGHT-kg						
POST-OPERATIVE DAY						
DATE						
TIME						
TEMPERATURE	40					
	39					
	38					
	37					
	36					
	35					
RESPIRATORY RATE						
O_2 SATURATION						
BLOOD PRESSURE AND PULSE RATE PULSE = X SYSTOLIC = V DIASTOLIC = ^	240					
	220					
	200					
	180					
	160					
	140					
	120					
	100					
	80					
	60					
	40					
	20					
BP						
PAIN SCORE SCALE						
INITIALS AND JOB NUMBER						

0 000000 001939

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MRN: | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

PHYSICIAN ORDER FORM

ALLERGIC TO:

DATE	TIME	PHYSICIAN ORDERS	PHYSICIAN STAMP / SIGNATURE	RN INITIAL / JOB NUMBER

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MRN: | | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NURSES PROGRESS NOTES FORM

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

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